# THE LCI-LOCUS HILBERT STACK IS FORMALLY SMOOTH ON OBJECTS

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*Editorial note.* No omission markers in-span; proof complete as written.

*Allowed standard prerequisites (editorial).*

- Unobstructed lifting. lemma-lci-unobstructed (same chapter, named). Provides the lifted Z'.
- Open substack. lemma-lci-locus-stack-in-groupoids (same chapter, named). Certifies the lift stays in the LCI locus.

*Source reference.* Original equation 06D0: sources/stacks-criteria/pinned-original-source.tex, source lines 886–886. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original equation 06CK: sources/stacks-criteria/pinned-original-source.tex, source lines 2744–2744. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 06CL: sources/stacks-criteria/pinned-original-source.tex, source lines 2785–2785. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 06D8: sources/stacks-criteria/pinned-original-source.tex, source lines 2836–2836. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $F : \mathcal{X} \to \mathcal{Y}$ be a 1-morphism of stacks in groupoids over $(Sch/S)_{fppf}$. Assume $F$ is representable by algebraic spaces, flat, and locally of finite presentation. Then*

$$p : \mathcal{H}_{d,lci}(\mathcal{X}/\mathcal{Y}) \to \mathcal{Y}$$

*is formally smooth on objects.*

*Proof.* We have to show the following: Given

(1) an object $(U, Z, y, x, \alpha)$ of $\mathcal{H}_{d,lci}(\mathcal{X}/\mathcal{Y})$ over an affine scheme $U$,
(2) a first order thickening $U \subset U'$, and

*Date*: stacks-project:criteria.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(3) an object $y'$ of $\mathcal{Y}$ over $U'$ such that $y'|_U = y$,

then there exists an object $(U', Z', y', x', \alpha')$ of $\mathcal{H}_{d,lci}(\mathcal{X}/\mathcal{Y})$ over $U'$ with $Z = U \times_{U'} Z'$, with $x = x'|_Z$, and with $\alpha = \alpha'|_U$. Namely, the last two equalities will take care of the commutativity of (06D0).

Consider the morphism $x_\alpha : Z \to X_y$ constructed in Equation (06CK). Denote similarly $X'_{y'}$ the algebraic space over $U'$ representing the 2-fibre product $(Sch/U')_{fppf} \times_{y',\mathcal{Y},F} \mathcal{X}$. By assumption the morphism $X'_{y'} \to U'$ is flat (and locally of finite presentation). As $y'|_U = y$ we see that $X_y = U \times_{U'} X'_{y'}$. Hence we may apply Lemma 06D8 to find $Z' \to U'$ finite locally free of degree $d$ with $Z = U \times_{U'} Z'$ and with $Z' \to X'_{y'}$ extending $x_\alpha$. By construction the morphism $Z' \to X'_{y'}$ corresponds to a pair $(x', \alpha')$. It is clear that $(U', Z', y', x', \alpha')$ is an object of $\mathcal{H}_d(\mathcal{X}/\mathcal{Y})$ over $U'$ with $Z = U \times_{U'} Z'$, with $x = x'|_Z$, and with $\alpha = \alpha'|_U$. As we've seen in Lemma 06CL that $\mathcal{H}_{d,lci}(\mathcal{X}/\mathcal{Y}) \subset \mathcal{H}_d(\mathcal{X}/\mathcal{Y})$ is an "open substack" it follows that $(U', Z', y', x', \alpha')$ is an object of $\mathcal{H}_{d,lci}(\mathcal{X}/\mathcal{Y})$ as desired. $\square$